Practical 7

Aim : Study about Hyperledger Composer.

Introduction :

Hyperledger Composer was an open-source development framework used to develop blockchain business networks. It was part of the Hyperledger project and was designed to work with Hyperledger Fabric.

Composer provided a simple way to define the business logic of a blockchain application. Developers could define participants, assets, transactions, and business rules without directly handling all the complex technical details of the blockchain network.

It was mainly designed for permissioned blockchain networks, where participants are known and have specific permissions.

Main Components :

1. Participants

Participants represent individuals, organizations, or entities involved in a business network.

Examples: Buyer, Seller, Bank, Supplier.

2. Assets

Assets represent objects or resources that have value and are managed on the blockchain.

Examples: Vehicle, Property, Product, Certificate.

3. Transactions

Transactions are operations that change the state of assets or the business network.

Examples: TransferVehicle, BuyProduct, RegisterProperty.

4. Queries

Queries are used to search and retrieve information stored in the blockchain network.

5. Events

Events are generated when a particular transaction or activity occurs. They allow applications to receive notifications about changes in the network.

6. Access Control

Access control defines what different participants are allowed to do in the business network. This ensures that only authorized users can perform specific operations.

Important Files :

File	Purpose
.cto	Defines participants, assets and transactions
.js	Contains transaction-processing logic
.acl	Defines access permissions
.qry	Defines queries

Example :

Consider a Vehicle Registration System developed using Hyperledger Composer.

Participants:

- Vehicle Owner
- Dealer
- Government Authority

Asset:

- Vehicle

Transactions:

- Register Vehicle
- Transfer Vehicle

For example, when a vehicle is sold, a Transfer Vehicle transaction can change the ownership of the vehicle from one participant to another.

Features

- Simplifies blockchain application development.
- Provides business network modeling.
- Supports participants, assets, and transactions.
- Provides access-control mechanisms.
- Supports queries and events.
- Integrates with Hyperledger Fabric.

Advantages

1. Easy to understand and use for blockchain application development.
2. Reduces the amount of low-level blockchain programming.
3. Provides a structured way to represent business processes.
4. Useful for permissioned blockchain networks.
5. Provides APIs for connecting applications with the blockchain network.

Applications

Hyperledger Composer was used for applications such as:

- Supply chain management
- Vehicle registration
- Healthcare records
- Banking and finance
- Property management
- Product and asset tracking

Limitation

Hyperledger Composer was deprecated and is no longer actively maintained. It was mainly used with earlier Hyperledger Fabric development workflows. Modern Hyperledger Fabric applications generally use current Fabric development tools instead.

Conclusion

Thus, Hyperledger Composer, its basic components, important files, features, advantages, applications, and limitations were studied and understood.